

Opening Reflections: The Evolving Ocean

Thank you

Organisers gave me quite some freedom - dangerous

Would like to cover a range

Ocean

REN21

Framing

National governance

Marine Spatial Planning

Environmental Impact Assessment

Sidenote: deep-sea mining

1. Framing the Challenge: Crowded Seas, Fragmented Governance

Industrialisation

EU Blue Economy – biotechnology, tourism, aquaculture

- **Rapid offshore energy growth:**

- We're witnessing an acceleration in offshore wind farms, undersea transmission cables, floating hydrogen hubs, carbon storage facilities- and even solar arrays at sea.
- Offshore wind capacity surged globally, reaching 75.2 GW in 2023 (+10.8 GW).
- China led with 6.3 GW added; Europe (UK, Germany, Denmark) remain critical hubs.
- EU's ambitious targets: 76 GW by 2030, 300 GW by 2050 (NSEC).

- **Ocean power lags far behind:**

- Just 530 MW installed globally, with minimal growth (~1.5 MW in 2023).
- Persistent barriers: high costs, technology challenges, fragmented legal support.

- **Legal tensions from rapid transition:**

- Marine spaces increasingly contested, infrastructure straddles sectors and boundaries.
- New governance tools – MSP – promising, but also increasingly under the spotlight
- International law struggles to keep pace, creating legal ambiguity and political tensions.
- Need to consider new governance paradigms—“**blue legalities**”

2. Examples of Governance Fragmentation

National

Sectoral governance

Ocean Power: Scotland's Governance Case Study

- Scotland's MSP perceived as international best practice but industry views it as fragmented, bureaucratic, unpredictable:
 - Developers face multi-agency approvals, unclear tenure, lengthy delays.

- Scotland's model highlights tension: ambitious rhetoric vs. cumbersome implementation.

Offshore Wind: North Sea Interconnections

- Who governs shared grids? Who pays, controls, and benefits?
- **Cross-border infrastructure (e.g., North Sea Energy Cooperation, North Sea Wind Power Hub):**
 - Who governs shared grids? Who pays, controls, and benefits?
 - Existing bilateral agreements (e.g., Denmark-Belgium energy islands, UK-Norway cables) test sovereignty limits and highlight gaps in legal frameworks.
 - EU's MSP Directive mandates national plans but offers weak enforcement across borders.

2. The Challenge of Marine Spatial Planning (MSP)

MSP – process to understand conditions, future developments, develop a plan through political process

EU Directive

Highly technical process.

Promoted extensively by UNESCO and others – logical

Academic – increasing concern and critique

- **MSP: intended as integrative governance, often reinforces sectoral dominance:**
 - Shipping, fishing, and energy sectors dominate planning processes.
 - Legal scholars argue MSP reinforces state sovereignty, limiting genuine cross-border cooperation.
- **Political economy of MSP:**
 - Offshore wind – industry may feel it can't compete with existing users – existing user may feel wind gets unfair prioritization, simplified licensing—embedding a techno-economic bias.
 - Civil society, coastal communities, and traditional marine users often marginalized, leading to procedural injustice.
 - Or overconsulted – Orkney
- **MSP critiques:**
 - Risk of "strategic tokenism": stakeholder participation superficial rather than genuine deliberation.
 - Need to reorient MSP to include adaptive, justice-based governance frameworks (e.g., giving coastal communities meaningful decision-making roles).

Marine spatial planning is indispensable yet insufficient without legal coherence, adaptive governance, and political will.

- As offshore energy expands, MSP must evolve from sector conflict mediation to governance for transformation — enabling just, inclusive, and science-informed transitions.
- Rather than asking whether MSP works, we must ask: for whom, under what vision of sustainability, and with what enforceable limits?

3. Science-Policy Interface: EIAs and Adaptive Governance

Impacts – can mitigate - RESR

- **Environmental Impact Assessments (EIAs):**
 - Often reduced to bureaucratic compliance rather than dynamic governance tools.
 - Unclear thresholds for significant impacts create regulatory confusion, excessive caution, or gridlock.
- **Scientific complexity vs. regulatory simplicity:**
 - Marine ecosystems present high uncertainty and cumulative impacts not easily captured in EIAs.
 - Science frequently mobilized strategically in political disputes—NGOs vs. developers—rather than as impartial evidence base.
- **Case for adaptive management:**
 - Need legal frameworks embedding adaptive, iterative EIAs (monitor-adjust-decide loops).
 - Promote shared data platforms, clear thresholds calibrated by deployment scale.

5. Deep Seabed Mining (DSM): Lessons from an Emerging Legal Crisis

- **Governance gaps exposed by DSM:**
 - ISA's delayed "Mining Code" and unresolved legal frameworks showcase institutional inertia.
 - Divergence within EU (France, Germany support moratorium; Belgium, Poland have mining interests).
- **Environmental and geopolitical risks:**
 - DSM presents irreversible ecological threats, raising ethical and generational equity issues.
 - Highlights legal fragmentation in managing new marine industries with high ecological risks.

6. Towards a New Legal Paradigm: Integrative Ocean Governance

- **Integrative governance principles:**
 - Precautionary principle and intergenerational justice as fundamental legal anchors.
 - Adaptive, cross-border governance structures, bridging sectoral gaps and ecological limits.
- **EU and regional frameworks as potential models:**

- North Sea Energy Cooperation (NSEC) and EU Green Deal show emerging regional cooperative models.
- Yet, these models must strengthen enforcement and integrate ecological justice deeper into legal frameworks.
- **Recommendations for reform:**
 - Harmonize cross-border legal frameworks, explicitly addressing infrastructure ownership and liability.
 - Strengthen participatory processes, incorporating customary marine rights, local community deliberation, and transparent science-based decision-making.
 - Foster interdisciplinary approaches: legal, ecological, political sciences must jointly address complex governance issues.

Strategic Reflections & Key Messages (Conclusion)

- **Marine renewables reveal a critical legal-political shift:**
 - The rapid energy transition outpaces current maritime governance capacities.
 - Fragmented law risks ecological degradation and socio-economic injustice.
- **Transnational cooperation essential:**
 - Bilateral and regional legal instruments require harmonization to manage shared ocean spaces effectively.
- **Future-proofing marine governance:**
 - Legal frameworks must evolve rapidly to embrace adaptive, integrative, and just ocean stewardship principles, anchored in rigorous science and genuine stakeholder engagements